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SECP3744 ENTERPRISE SYSTEMS DESIGN AND MODELING (WBL)

SECTION 01

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Introduction

In the contemporary digital landscape, higher education institutions increasingly depend on integrated information systems to augment operational efficiency, increase service delivery, and foster academic achievement. Universiti Tun Hussein Onn Malaysia (UTHM) employs a systematic Enterprise Architecture (EA) methodology to synchronise its technology infrastructure with institutional objectives. EA offers a strategic framework that integrates individuals, processes, and technology, guaranteeing the coherent operation of diverse systems to facilitate administrative, academic, and research activities.

At UTHM, this architectural strategy is manifested through the integration of various essential subsystems that enhance daily operations for students and staff alike. The CENTRAL Portal and Housing Office Management System (HOMS) are essential for providing information access, administering student services, and enhancing operational efficiency. Endorsed by the university's Information Technology Centre (PTM), these subsystems illustrate how meticulously designed architecture improves user experience and institutional efficacy.

UTHM employs a variety of technologies including programming languages, databases, frameworks, and emerging innovations to fulfil academic, administrative, and research requirements. UTHM exemplifies a dedication to digital transformation, sustainability, and continuous improvement by merging contemporary approaches such as SOA and TOGAF with advanced technologies including AI, IoT, blockchain, and cloud computing.

Enterprise Architecture (EA)

An enterprise architecture (EA) is a conceptual blueprint that defines an organization's structure and operation. Its intent is to determine how an organization can effectively achieve its current and future objectives (Barney & Gillis, 2025). Enterprise architecture involves analyzing, planning, designing and eventually implementing the analysis in an enterprise. Its use helps businesses going through digital transformation, since EA focuses on bringing legacy applications and processes together to form a seamless environment.

The existing literature on Enterprise Architecture (EA) in higher education highlights its vital role in aligning an institution's Information Technology (IT) with its strategic business goals, moving beyond simple system development to address integration and efficiency concerns

(Amalia & Supriadi, 2017). A common practice in this field is the adoption of standardized frameworks, most notably TOGAF (The Open Group Architecture Framework), which is favored for its detailed methodology and flexibility in integrating various information units and systems across different standards and platforms. Studies such as the one by Amalia and Supriadi represent a key contribution to this research area, specifically by demonstrating the application of the TOGAF Architecture Development Method (ADM) to generate a practical blueprint, identify necessary application candidates, and establish the governing principles for an integrated IT architecture in a university setting. This Figure 2.1 below shows a framework of its EA.

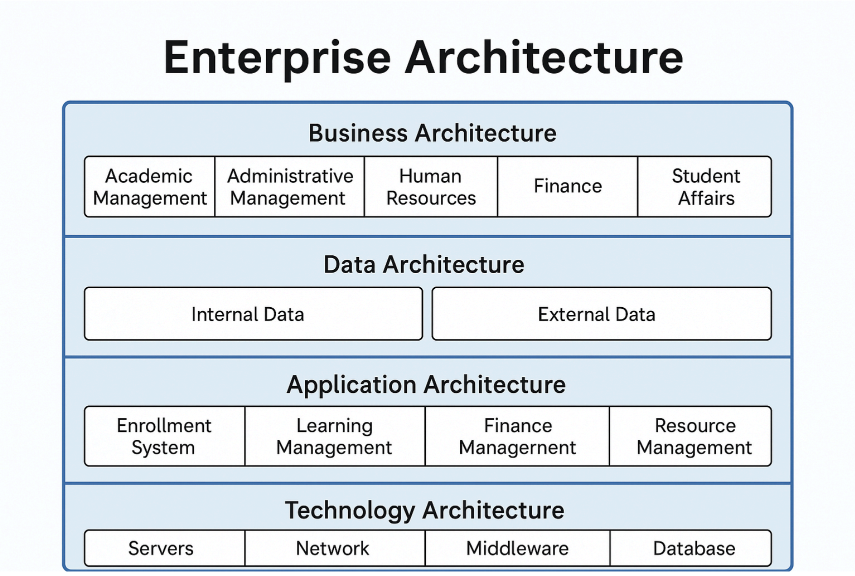

Figure 2.1 Enterprise Architecture Overview

Related Subsystems

Enterprise Architecture (EA) in higher education institutions typically integrates multiple subsystems that support academic, research, governance, administration, and community functions. These subsystems interact across different layers, including business processes, service delivery, information technology platforms, and stakeholder interactions, in an attempt to align institutions with their strategic goals. Many universities adopt EA for simplifying services provided to students and staff, standardising information management, and increasing decision-making effectiveness (Amin et al., 2024).

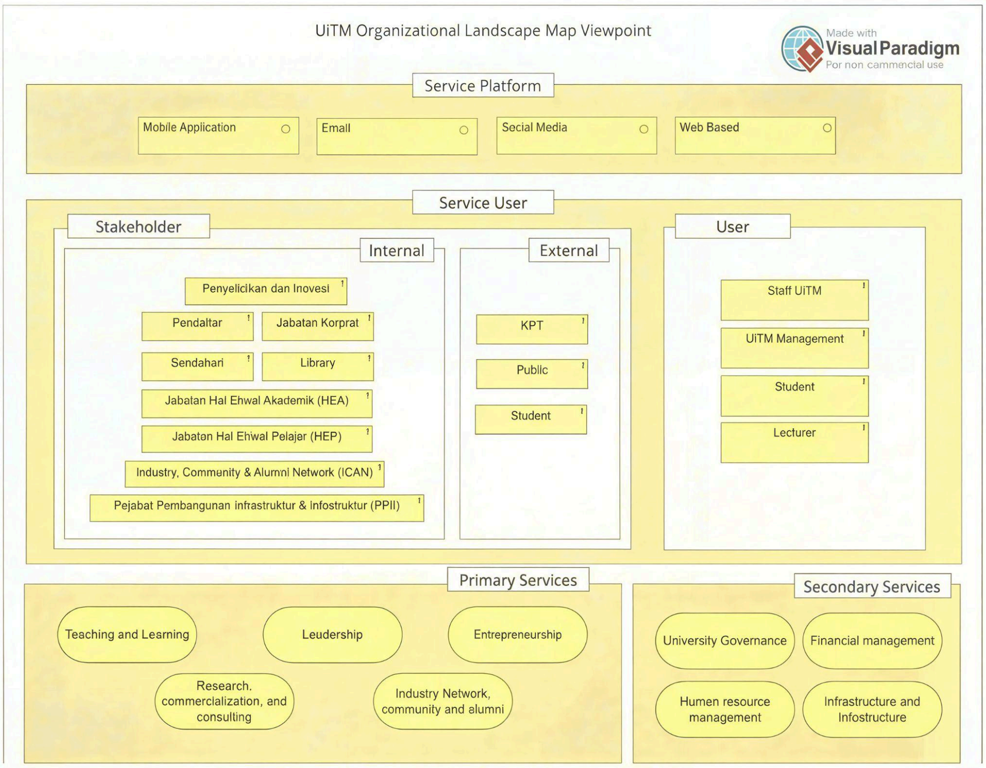

Figure 2.2 UiTM Organizational Landscape Map Viewpoint

Figure 2.2 shows UiTM's Enterprise Architecture (EA) landscape, which is built around different subsystems. Each subsystem handles specific tasks, such as academic services, administrative services, or digital platforms, and they all work together to connect stakeholders and service users. The framework includes internal and external stakeholders, main and supporting service areas, and works across platforms like mobile, email, social media, and websites. By linking these subsystems, UiTM creates a smooth, integrated service environment that supports its goals for improving the institution (Amin et al., 2024).

Subsystem Category	Subsystem	Description
Service Platforms	Mobile App, Email, Social Media, Web-based Systems	Digital channels enabling communication and service access for users.
Stakeholders	Internal: HEA, HEP, Library, Corporate, ICAN, PPII, etc.	Organizational units involved in academic, student, infrastructure, alumni, and corporate services.
	External: KPT, Public, External Students	External bodies influencing or consuming UiTM services.

User Groups	UiTM Staff, Management, Students, Lecturers	Primary service users interacting with UiTM digital and administrative services.
Primary Services	Teaching & Learning, Leadership, Entrepreneurship, Research & Commercialization, Alumni & Industry Network	Core mission-driven services supporting UiTM's academic and developmental goals.
Secondary Services	University Governance, Human Resource, Finance, Infrastructure	Supporting subsystems enabling operations, compliance, and resource management.

Table 2.1 Description of UiTM EA Subsystems

Table 2.1 provides an overview of how each subsystem at UiTM supports the services offered at the university and the involvement of different stakeholders. It categorises subsystems into functional types, describes their purpose, and identifies service platforms on which users can access them. It also provides information on the internal and external stakeholders and user groups, and the primary and secondary services each subsystem enables. This structured view shows how UiTM integrates its digital and administrative systems into a seamless, mission-driven service environment.

Feature	**Layered EA Model**	**UiTM Organizational Landscape Map Viewpoint**
Main Focus	**Architectural Components** (What are the components of the enterprise's IT?).	**Organizational and Service Components** (Who uses the services and what services are offered?).
Business/Organizational Content	Lists **Functional Areas**: Academic Management, Finance, Human Resources, etc.	Lists **Key Stakeholders** (Internal/External), **Users** (Staff, Student, Lecturer), and **Services** (Teaching, Leadership, Governance).

IT/Technology Content	Lists **Abstract Domains**: Servers, Network, Middleware, Database (under Technology Architecture).	Lists **Delivery Channels/Platforms**: Mobile Application, Email, Social Media, Web Based (under Service Platform).
Framework Link	Strongly aligned with the **TOGAF/Zachman** frameworks' core layered structure.	Represents a **Viewpoint** or business-centric model, often part of an overall EA deliverable.

Table 2.2 Comparison between Layered EA Model and UiTM Organizational Landscape Map Viewpoint

This Table 2.2 visual two representations of Enterprise Architecture (EA) serve distinct but complementary roles: the Layered EA Model acts as a structural blueprint, primarily focusing on Architectural Components by detailing the required Functional Areas like Academic Management and the underlying abstract IT Domains such as Servers and Network, consistent with the formal, tiered structure of frameworks like TOGAF; conversely, the UiTM Organizational Landscape Map Viewpoint functions as a high-level, business-centric communication tool, focusing on Organizational and Service Components by illustrating the key Stakeholders (Staff, Students) and the Delivery Channels (Mobile Apps, Web) they use to access the university's services.

Technology Used

SAP Business Technology Platform (SAP BTP) is SAP's unified cloud platform designed to support the development, integration, and extension of enterprise applications. It provides a comprehensive suite of services that enable organizations to manage data, build applications, automate business processes, and integrate systems in a scalable and secure environment.

Core Components of SAP BTP

1. **Database and Data Management**
   Database and data management focuses on efficient data storage, processing, and governance. Key services include SAP HANA Cloud, SAP Data Warehouse Cloud, and SAP Data Intelligence, which collectively support real-time data access and

enterprise-level data management.

2. **Analytics**
   SAP BTP includes analytical tools such as SAP Analytics Cloud (SAC), enabling organizations to perform reporting, data visualization, predictive analytics, and planning. These tools help businesses make informed, data-driven decisions.

3. **Application Development and Automation**
   This component offers capabilities for building and extending applications using low-code/no-code tools, cloud-native development frameworks, and process automation. Tools such as SAP Build Apps, SAP Build Process Automation, and the SAP Cloud Application Programming Model (CAP) streamline application development and workflow automation.

4. **Integration**
   SAP BTP supports seamless connectivity between SAP and non-SAP systems through services like SAP Integration Suite, API Management, and Event Mesh. These tools ensure smooth data exchange and interoperability across the organization's technology landscape.

Importance of SAP BTP

SAP BTP enables organizations to modernize business processes, reduce dependency on custom code in core systems, and create flexible, scalable solutions. By unifying application development, data management, analytics, and integration services, the platform helps businesses enhance innovation, improve efficiency, and adapt quickly to changing operational needs.

References

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